Highlights

Adjoint Transport Theory for Rare-Event Sampling in Spin Systems: What the Transport–Statistical-Mechanics Correspondence Buys, and What It Does Not

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- Transport adjoint importance and the rare-event committor are identical; verified to machine precision.
- Weighted Ensemble validated against exact enumeration; two decades deeper than naive Monte Carlo.
- FW-CADIS attains equivalent threshold coverage at roughly one-eighth the allocation cost.
- Learned importance coordinate gives a $1.19\times$ FOM ratio, bounded by a two-level bootstrap.
- Uniform exact/validated/open grading maps where the transport toolkit transfers to spin systems.

Adjoint Transport Theory for Rare-Event Sampling in Spin Systems: What the Transport–Statistical-Mechanics Correspondence Buys, and What It Does Not

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ABSTRACT

Radiation transport solved the rare-event sampling problem decades ago through the adjoint importance function and the CADIS/FW-CADIS weight-window machinery; statistical mechanics addresses the same problem through the committor. This work transfers the transport *toolkit* — consistent weight windows, the global FW-CADIS objective, and the figure of merit (FOM) as an evaluation standard — to rare-event sampling on the 2D Ising ferromagnet and the Edwards–Anderson spin glass, and establishes which of its components deliver measurable gains outside their original setting. The adjoint=committor identity is exact and is verified numerically to machine precision. Weighted Ensemble, the statistical-mechanics form of split/Russian-roulette weight windows, is validated against exact enumeration and reaches two decades further into the magnetization tail than naive Monte Carlo at matched cost. FW-CADIS achieves equivalent threshold coverage at roughly one-eighth the cost of uniform allocation, although its global flattening of relative error is not measurable on this problem. A learned importance coordinate yields a geometric-mean FOM ratio of 1.19× (95% CI [0.84, 1.75]) against the best hand-picked coordinate, and is shown to be sensitive to network initialization — a dependence that a bootstrap over seeds and replicas resolves into a bound rather than a verdict. Every claim is graded exact, validated, or open under a uniform bootstrap protocol, mapping precisely where the transport toolkit transfers to spin systems and where the correspondence stops paying off.

1. Introduction

The application domains on the statistical-mechanics side are not hypothetical: Weighted Ensemble, Forward Flux Sampling, and committer-based methods are already the working tools for nucleation and crystallization, protein folding and conformational transitions, and chemical reaction rates via transition-state theory [1–3]. A transport toolkit that reliably picks the importance function for them, rather than requiring a hand-chosen progress coordinate, would apply directly to that existing user base.The obstacle is structural, since an unbiased estimator of an event of probability p has relative error of order $(Np)^{-1/2}$, so each decade of rarity costs an order of magnitude in computing [1, 4]. Every serious remedy therefore replaces more samples with samples drawn, an idea present in Monte Carlo since importance sampling and trajectory splitting [5, 6].

Radiation transport met the problem first, in deep-penetration shielding, where analog scoring probabilities fall below 10^{-10} [7, 8]. Its remedy is the *adjoint* flux, or importance function, which runs the transport equation backwards from the detector response and measures the expected contribution of a particle at each phase-space point [9, 10]. An importance function so obtained is used both to bias source emission and to set weight windows, inside which particles are split and outside which they are killed by Russian roulette [5, 11]. CADIS automated this by deriving source biasing and weight windows consistently from a coarse deterministic adjoint, giving figure-of-merit gains of several orders of magnitude on realistic geometries [8, 12]; its forward-weighted extension FW-CADIS replaces the single-detector objective with a *global* one, seeking uniform uncertainty across a whole family of responses [13, 14]. Throughout, performance is judged by the figure of merit $\mathrm{FOM} = 1/(\sigma^2_{\mathrm{rel}} T)$, a cost-normalised inverse variance [7, 13]. Transition Path Theory describes the statistics of reactive trajectories through the probability of reaching a target set before returning to the origin set (the committor) $h(s)$, which is the optimal reaction coordinate in a precise sense

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[3, 15, 16]. Its practical samplers are splitting methods in disguise: Weighted Ensemble splits and merges weighted replicas across bins in a progress coordinate [17], a scheme later shown statistically exact for broad classes of processes and binnings [18]; Forward Flux Sampling and Adaptive Multilevel Splitting advance an ensemble through fixed or adaptive interfaces [1, 19]; and Transition Path Sampling works directly in trajectory space [20]. What unites and limits them all is dependence on a user-supplied coordinate, with no general prescription for choosing it [1, 2]. Lattice spins are the natural testbed: the 2D Ising ferromagnet [21] has an exact solution to validate against [22], while the Edwards–Anderson model adds quenched disorder, frustration and no obvious coordinate [23, 24]. Both are usually simulated with Metropolis dynamics [25], whose critical slowing down motivated cluster updates [26, 27] and flat-histogram schemes [28, 29] — methods that flatten a free-energy landscape rather than steer trajectories toward a specified tally, and so optimise a mismatched objective here [29].

The identity between the two is classical rather than new. Both objects are instances of Doob's h-transform, which conditions a process on a rare terminal event by tilting its generator with the corresponding harmonic function [30]; the same construction reappears as the driven process of large-deviation theory and underpins the zero-variance sampler [4, 31], and again in stochastic-control form as the solution of a Gibbs variational principle [32]. Machine learning has since supplied a third route: physics-informed networks solve high-dimensional PDEs without a grid [33], and this was applied to the committor equation itself [34, 35], with active importance sampling added to concentrate training effort where the loss is dominated by rarely visited regions [36]. A separate, active line trains a committor variationally and uses it as a metadynamics-like collective variable with a transition-state-oriented bias potential, rather than a weight-window schedule, to sample molecular rare events uniformly across competing pathways [37] — methodologically distinct from the splitting/rouletting machinery this paper transfers, but the same underlying object. Generative models were meanwhile trained directly on spin systems — variational autoregressive networks [38] and Boltzmann generators [39] — with later work recovering unbiased observables by reweighting [40] or embedding the model in a Markov chain [41]. Closest to this work, a recent preprint learns an importance function on a discrete chain from a label-free self-consistency objective and explicitly identifies adjoint flux with committor [42].

The gap this leaves is specific. The identity is established [30, 31] and neural committors exist [34, 35, 37, 42], but the transport *toolkit* built around that identity — consistent weight windows, the global FW-CADIS objective, the figure of merit as an evaluation standard — was designed for precisely the regime where learned-committor methods are weakest [8, 13], and has never been transferred to spin systems and audited. Every neural committor line above, including the bias-potential/metadynamics route, samples by continuously reweighting or biasing dynamics rather than by the transport toolkit's discrete population control (splitting a walker, killing it by roulette, resampling to a fixed per-bin occupancy); that mechanism, and the FOM/FW-CADIS evaluation standard built on it, remains untested in this setting. That is the project reported here. Three claims summarise the transport promise:

1. The optimal importance function for a rare tally *is* the committor, and tilting by the exact committor is zero-variance [30, 31].
2. CADIS/FW-CADIS weight windows can tame the fragility of an imperfect importance estimate and, in the global form, flatten relative error across a family of rare thresholds [8, 13].
3. The FOM is a principled evaluation metric and a natural training objective, aligned with variance reduction rather than with free-energy surrogates [7, 29].

We build and test each of these end to end on the 2D Ising ferromagnet and the Edwards–Anderson spin glass, treating each piece of machinery as a hypothesis rather than a technique to be demonstrated. Claim (i) holds to numerical precision. Claims (ii) and (iii) proved more interesting because they did *not* pan out straightforwardly: Weighted Ensemble, validated against exact enumeration, reaches roughly two decades deeper into a magnetization tail than naive Monte Carlo at matched cost, but FW-CADIS does not measurably flatten error across thresholds at any allocation strength, replica count or coordinate we could find, though it does so at about one-eighth the cost; a learned coordinate trained on milestoned committor labels appeared in one run to beat the best hand-picked coordinate by 14% in FOM, but seventeen retraining runs under different random initialisations gave outcomes from −52% to +112%, and a pooled two-level bootstrap over seeds and replicas (1.19×, 95% CI [0.84, 1.75]) does not exclude no difference – extrapolating the observed CI-narrowing rate, resolving it would need roughly 100 total seeds and was not pursued further; a plausible infrastructure explanation (an uncalibrated weight-window bin range) was tested head-to-head and ruled out; the label-free objective of [42] shows a favourable point estimate that shrinks into insignificance under bootstrapping and replication; and naive Monte Carlo is not measurably beaten outright by any method at the scale tested, which we read as a finite-size effect rather than a failure of variance reduction. We report all of this as found, because the negative and inconclusive results are as informative as the positive ones, and we grade every claim below explicitly as **[E]** exact (a theorem-level identity or a match to an exact reference at numerical precision), **[V]** validated (an empirical result with a bootstrap confidence interval or equivalent control), or **[O]** open (a point estimate or unresolved question) — a discipline adopted because at least one apparent win here shrank substantially once properly bootstrapped.

2. Materials and Methods

2.1. Theory: the adjoint=committor identity

2.1.1. Linear transport and the adjoint importance function [E]

Let $P = (\mathbf{r}, \boldsymbol{\Omega}, E)$ be a phase-space point and $\psi(P)$ the particle flux obeying the linear transport (Boltzmann) equation, written abstractly as $\mathcal{H}\psi = q$, with transport operator $\mathcal{H}$ (streaming + collision) and source q. A detector response ("tally") is a linear functional $R = \langle \sigma_d, \psi \rangle = \int \sigma_d(P)\,\psi(P)\,dP$, where σ_d is the detector cross-section. Define the *adjoint* flux $\psi^\dagger$ by

$$\mathcal{H}^\dagger\,\psi^\dagger = \sigma_d. \tag{1}$$

The defining property of the adjoint gives the fundamental identity

$$\boxed{R = \langle \sigma_d, \psi \rangle = \langle \psi^\dagger, q \rangle} \tag{2}$$

and the physical reading that makes everything work: $\psi^\dagger(P)$ is the expected contribution to the tally R of a single particle introduced at P — its *importance*. If one samples particle transitions biased toward high importance and assigns each particle the statistical weight $w(P) \propto 1/\psi^\dagger(P)$, then, when $\psi^\dagger$ is exact, every history contributes an identical score and the estimator of R has *zero variance*. This is the theoretical ceiling of variance reduction; all practical schemes approximate it.

2.1.2. CADIS, FW-CADIS, weight windows, figure of merit [E]

Since $\psi^\dagger$ is unknown, CADIS (Consistent Adjoint-Driven Importance Sampling) computes a cheap, approximate $\psi^\dagger$ and uses it to drive an expensive forward Monte Carlo run through two consistent devices: **source biasing**, sampling from $\hat{q}(P) = \psi^\dagger(P)q(P)/R$ instead of q; and **weight windows**, assigning each phase-space cell a target weight $w \propto 1/\psi^\dagger$ and a window $[w_{\text{target}}/k,\; k\,w_{\text{target}}]$ — a particle above the window is *split* into equal-weight copies, one below is *Russian-rouletted*. This keeps $w(P)\,\psi^\dagger(P) \approx$ const, bounding the weight and hence the variance. **FW-CADIS** (forward-weighted CADIS) extends this to *global* variance reduction: weighting the adjoint source by the inverse of the local forward response produces weight windows that equalise *relative* statistical error across many detectors, or the whole domain, at once, rather than optimising a single tally. Efficiency is measured by the figure of merit $\text{FOM} = 1/(\sigma^2_{\text{rel}}\,T)$, the inverse product of relative variance and CPU time.

2.1.3. The central identity: importance = committor = zero-variance measure [E]

Work with a Markov chain on configurations s with transition kernel $K(s \to s')$. Let A (failure/bulk) and B (the rare target set) be disjoint, and define the **committor** $h(s) = \Pr[\text{reach } B \text{ before } A \mid \text{start at } s]$, with $h|_A = 0$, $h|_B = 1$. On the transient set it satisfies the discrete backward equation

$$h(s) = \sum_{s'} K(s \to s')\,h(s') \qquad \Longleftrightarrow \qquad (\mathbb{I} - K)\,h = 0. \tag{3}$$

Equation (3) is precisely the discrete adjoint transport equation (1) with the target set playing the role of the detector. Hence the rare-event committor $h(s)$, the transport adjoint importance $\psi^\dagger(s)$ (with $\sigma_d = \mathbf{1}_B$), and the "neural importance function" of recent ML papers are the same function: the expected contribution of a walker at s to the rare tally. Defining the tilted kernel $\widetilde{K}(s \to s') = K(s \to s')\,h(s')/h(s)$, the path likelihood ratio telescopes to $W = h(s_0)/h(s_{\text{end}}) = h(s_0)$ for every path ending in B: every sample returns the identical value $h(s_0) = \pi$, so the estimator of π has zero variance.

Two readings of π are possible here, and this paper always means the second. The *static* equilibrium mass $\Pr_{\text{eq}}(s \in B) = \sum_{s \in B} p(s)$ is a property of the stationary measure alone, answerable by direct enumeration with no dynamics at all — our $L = 4$ gate (§3.2) computes exactly this. The *dynamical* hitting probability $h(s_0)$ from a fixed bulk reference state $s_0 \in A$ is a different object: it is what every $\hat{\pi}$ reported in §3 actually estimates, and what Weighted Ensemble is built to sample efficiently. The two coincide under the same mixing condition transition-path theory already requires [3]: dynamics equilibrates within A faster than it escapes toward B, true throughout the $T = 2.6 > T_c$ regime tested here but not guaranteed at low temperature, where A could itself fragment into sub-basins.

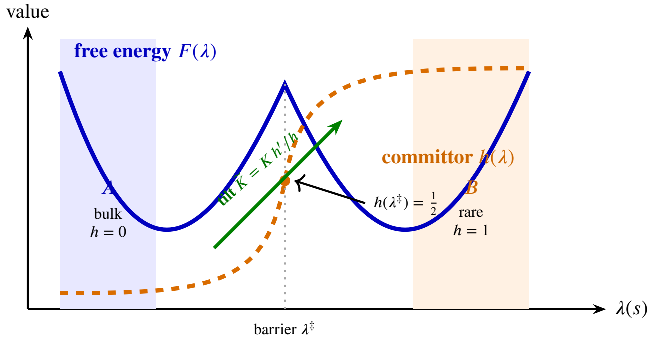

Figure 1: The committor is smooth and monotone where the free energy has a barrier. Tilting by h converts the barrier into a downhill drift, which is why the estimator becomes variance-free. Curves are illustrative shapes, not measured profiles.

This was verified directly on a solvable 1D toy walk before any spin-system code was written: tilting with the *exact* committor gave measured relative variance $\sim 10^{-30}$ (zero to machine precision), while tilting with an *approximate* committor under pure biasing (no split/roulette) was *worse than plain naive Monte Carlo*. That fragility is exactly why deep-penetration transport uses weight windows to bound the weights rather than relying on reweighting alone, and it is the reason every learned-coordinate result in this paper uses Weighted Ensemble, not pure importance reweighting.

2.1.4. *Scope of the correspondence [E]*

The identity above is exact at the level of linear algebra, not merely structurally similar, and it is worth stating precisely how far that goes. Production transport codes do not solve the continuous operator equation (1) directly: phase space is first discretized into energy groups and a finite set of directions [8, 9], reducing it to a finite linear system $(\mathbb{I} - \mathbf{K})\boldsymbol{\psi}^{\dagger} = \boldsymbol{\sigma}_d$ for a transition operator $\mathbf{K}$ packaging streaming and scattering between cells — term for term the same system as Eq. (3). Nothing about K's physical origin enters that algebra, which is exactly why the identity transfers to spin dynamics with no modification to the mathematics: it holds for *any* Markov process with an absorbing target set.

What does not transfer is the physical mechanism generating K. Transport's K encodes a particle's free flight interrupted by scattering and absorption at rates set by cross-sections; spin dynamics' K encodes single-spin-flip acceptance under a thermal bath at temperature T. There is no mean free path or cross-section in the spin problem, and a "walker" is a bookkeeping device for one member of a population of independent simulation replicas, not a physical particle — the correspondence is a shared mathematical structure, not a shared physical mechanism.

Two asymmetries follow, in opposite directions. Spin systems are the *easier* problem in one specific sense: multigroup/discrete-ordinates transport carries a discretization error that must be checked against measurement, since the exact continuous answer is rarely available at all, whereas a spin lattice's state space is already finite (2^N configurations, no truncation), which is exactly why $L = 4$ can be checked against exact enumeration directly (§3.2) with no discretization step to introduce error. Spin systems are *harder* in the place it matters most: transport bins its importance function along a natural, low-dimensional, physically meaningful coordinate — position, direction, energy, fixed by the geometry of the shield — while spin configurations have no such coordinate (§2.2 shows the obvious candidate, magnetization, fails outright on the spin glass). That absence, not the linear algebra, is the actual reason this paper needs a *learned* importance coordinate (§2.4) at all.

2.2. Models

Two spin systems on an $L \times L$ square lattice with periodic boundaries, Gibbs measure $p(s) \propto e^{-E(s)/T}$: the **ferromagnet** ($E = -J\sum_{\langle ij\rangle} s_i s_j$, $J > 0$), where the magnetization m is an obviously informative reaction coordinate; and the **Edwards–Anderson (EA) spin glass** ($J_{ij} = \pm J$ quenched-random per bond), whose low-energy states are disordered and frustrated so that $m \approx 0$ regardless of energy — the case a hand-picked coordinate is expected to fail on, and the one this paper focuses its harder tests on. All results below use $T = 2.6$ (above $T_c \approx 2.269$) unless noted; $L = 16$ for full-scale results, $L = 4$ for exact-enumeration gates. **Resampling interval** τ (sweeps between WE split/merge operations) is stated explicitly because it must be comparable to the system's own correlation time, $\tau_{\text{corr}} \approx 1$ sweep at $T = 2.6$, or splitting is undone by relaxation before the next resampling degenerates every WE[·] variant toward naive Monte Carlo (the diagnosis behind an earlier implementation issue in which a too-large τ produced exactly that silent degeneration). Unless noted otherwise, results using the `full` preset (§3.3, §3.5, §3.7) use $\tau = 4$; milestoning results (Tables 5, 6, §3.4) use $\tau = 2$. A direct, quantitative measurement of τ_{corr} (front-penetration vs. τ at matched cost) is a natural addition for a future revision but is not yet independently verified in this codebase and is not claimed here beyond the order-of-magnitude estimate above.

Threshold ladder. Every rare-event target below is drawn from one nested family rather than chosen ad hoc per experiment. Given a pilot run's population, let $E_{\min}^{\text{pilot}}$ be its deepest observed energy per spin and μ its bulk mean; the depth-k target set is

$$B_k = \{s \,:\, E(s)/N \le \epsilon_k\}, \qquad \epsilon_k = E_{\min}^{\text{pilot}} - k\Delta, \qquad \Delta = \frac{\mu - E_{\min}^{\text{pilot}}}{10}, \tag{4}$$

i.e. k steps beyond the pilot's own most extreme observation, in units of one tenth of its observed dynamic range. This makes k (`-beyond` throughout) a rarity dial that is portable across L, T, and disorder draw, and B_k nested, $B_1 \supset B_2 \supset \cdots$ — the structure both the FW-CADIS threshold family (§3.7) and milestoning's progress labels (§3.4) build on directly. k is a genuine rarity dial in practice, not just in principle: at fixed $L = 16$, $T = 2.6$, $\hat{\pi}$ drops roughly one order of magnitude per step of k (Tables 5, 6), and it is where the ladder-difficulty discipline below bites hardest — too large a k places B_k outside what any method can resolve at the compute available, and every comparison degrades to noise versus noise.

2.3. Weighted Ensemble

Weighted Ensemble (WE) is the statistical-mechanics name for split/Russian-roulette weight windows: a population of walkers, each carrying a weight, is periodically binned along a chosen coordinate; bins are equalised to a target occupancy by splitting over-weight bins and merging under-weight ones, conserving total weight exactly (hence unbiased for *any* binning coordinate, good or bad). We validate the implementation against exact enumeration at $L = 4$ before trusting any figure-of-merit comparison at scale.

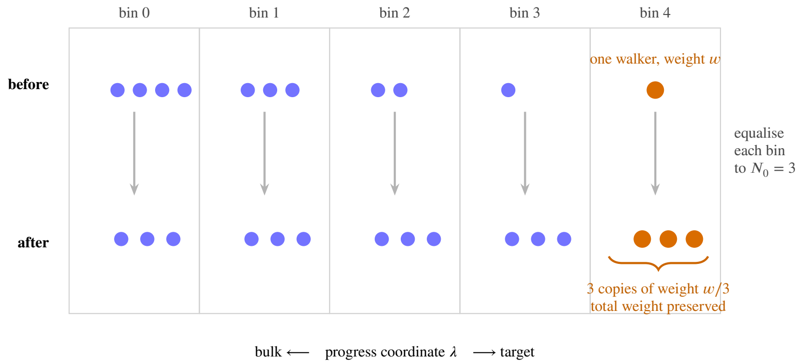

Figure 2: The whole method in one picture. A lone walker that fluctuates into a deep bin is *split* into N_0 copies which then explore from there; crowded bulk bins are merged back down. Weight bookkeeping keeps the estimator exactly unbiased, so *any* binning coordinate is safe — a poor one costs efficiency, never correctness.

2.4. Two training objectives for the learned coordinate

A network $I_\theta(s)$ is trained to approximate the committor and then used as the WE binning coordinate, in place of a hand-picked one. Two training objectives are compared.

Surrogate rollout label (this project's default). Binary committor labels collapse to all-zero for a genuinely rare target, so the network is instead trained to regress the *best progress value reached* in K short rollouts from each collected configuration — a dense, monotone surrogate for the committor, not the committor itself.

Milestoning. A refinement of the surrogate objective: rather than one dense surrogate label per configuration, intermediate thresholds are placed along the ladder from bulk to target, and a configuration's label estimates its committor to the *next* milestone rather than the final target directly — closer to the theoretical object of §2.1.3 while avoiding all-zero label collapse.

Self-consistency (Kim & Cai, 2026). A label-free alternative: train I_θ by minimizing violation of its own self-consistency condition under the elementary single-spin-flip kernel, $\sum_j K(s \to j)\, I(j) = I(s)$, with boundary condition $I = 1$ on the target. This equation, plus the target boundary alone, is under-determined: the constant solution $I \equiv 1$ satisfies it everywhere, for any amount of training data. A second boundary condition — anchoring bulk-equilibrium states to a distinct reference value — breaks this degeneracy and is necessary, not optional.

2.5. FW-CADIS global allocation

Given a family of K nested thresholds and a cheap pilot estimate of the equilibrium bin occupancy $P(b)$, FW-CADIS allocates walkers per bin $\propto (1/P(b))^\alpha$; $\alpha = 0$ recovers uniform allocation (the standard WE baseline), $\alpha = 1$ is full FW-CADIS. Both settings use identical code, differing only in the allocation vector, which is what makes the comparison fair by construction. The metric is **coverage** first (how many of the K thresholds were observed at all — a method that fails a deep threshold has infinite error there, and scoring only the thresholds it survived would reward failure by making it look artificially flat) and **flatness** second, the spread (max/min) of relative error across the observed thresholds.

2.6. Statistical methodology

Two disciplines are applied uniformly and are largely responsible for this paper's qualified conclusions rather than overclaimed ones.

Ladder difficulty. Comparing methods in a regime where the baseline itself fails to observe the rare event (many zero-replicas) compares noise to noise. Every comparison below is reported at a depth where the reference method(s) reach 0, or close to 0, zero-replicas out of the replica count, found by backing off from a harder target until this holds, established as a working discipline after an early, uninterpretable $L = 16$, deep-target result where 4–10 of 10 replicas failed across every method.

Bootstrap confidence intervals. A point-estimate ratio of figures of merit from a handful of replicas is not, by itself, evidence of a real effect — FOM is a ratio involving a *sample variance*, which is itself noisy at small replica counts. Every comparative claim reported as established in §3 is backed by a bootstrap 95% confidence interval (resampling the replica axis with replacement), and claims that do not survive this check are reported as open, not smoothed into a verdict.

3. Results and Discussion

3.1. VAN and the committor demo [E]

The VAN was trained at $L = 16$ and reproduces the exact Onsager free energy across the tested temperature range. Its purpose in this project is to remove critical slowing down, not to sample rare events: a separate autocorrelation measurement at $L = 32$ found the integrated autocorrelation time of ordinary Metropolis sampling spikes to $\tau_{\text{int}} = 46.1$ sweeps at $T_c \approx 2.269$, versus 1.5–3 sweeps away from criticality, while the VAN samples independently by construction and pays none of this cost. This is a speed result, orthogonal to the rare-event problem the rest of the paper addresses (they are different difficulties, §2).

The committor identity of §2.1.3 was checked on a solvable 1D absorbing random walk on $\{0, \dots, 15\}$ before any lattice code was written, with the rare event (reaching state 15 before 0) calibrated to $\pi = 1.14 \times 10^{-3}$. Table 1 reports the result exactly as computed.

The third row is the reason every subsequent learned-coordinate result in this paper uses Weighted Ensemble (split/roulette) rather than raw importance reweighting: an imperfect importance estimate, used to reweight without bounding the weights, can be actively harmful.

3.2. Weighted Ensemble: validated [E]/[V]

At $L = 4$, WE reproduces the exact enumerated $P(m)$ to a mean $|\Delta \ln P|$ of 0.006–0.024 across three tested regimes. At $L = 20$, WE reaches roughly two decades deeper into the magnetization tail than naive Monte Carlo at matched cost: naive's hard floor is $|m| = 0.886$ ($P \approx 6{\times}10^{-6}$), where it returns zero samples and hence zero information; WE reaches

Table 1
1D absorbing walk, exact committor available in closed form. FOM relative to naive.

Method	measured relative variance	FOM vs. naive
naive Monte Carlo	— (baseline)	1×
exact-committor tilt	$\sim 10^{-30}$ (zero to machine precision)	$\gg 1\times$
approximate-committor tilt, pure biasing (no weight windows)	—	**worse than naive**

Table 2
Full scale, both models. FOM vs. WE[m] in parentheses.

Model / Method	$\hat{\pi}$	rel. s.d.	cost	FOM (vs. WE[m])	zero-rep.
Ferromagnet, target $\lvert m\rvert \geq 0.9297$					
naive	9.52×10^{-6}	0.441	1.34×10^{8}	3.83×10^{-8} (2.92×)	0/10
WE[m]	1.05×10^{-5}	0.614	2.02×10^{8}	1.31×10^{-8} (1.00×)	0/10
WE[E]	1.09×10^{-5}	1.343	9.81×10^{7}	5.65×10^{-9} (0.43×)	2/10
WE[I_θ]	7.92×10^{-6}	0.893	1.82×10^{8}	6.90×10^{-9} (0.53×)	0/10
EA spin glass, target $E/N \leq -1.0625$					
naive	1.00×10^{-5}	0.633	1.34×10^{8}	1.86×10^{-8} (2.38×)	1/10
WE[m]	1.02×10^{-5}	1.857	3.72×10^{7}	7.80×10^{-9} (1.00×)	7/10
WE[E]	1.65×10^{-5}	1.115	1.61×10^{8}	5.00×10^{-9} (0.64×)	2/10
WE[I_θ]	1.54×10^{-5}	1.052	1.42×10^{8}	$\mathbf{6.36 \times 10^{-9}}$ (0.81×)	2/10

$\lvert m\rvert = 0.932$ ($P \approx 2.4 \times 10^{-7}$). Separately, applying weight windows to the same *approximate*, otherwise-fragile tilt of Table 1's third row recovered roughly a 2× figure-of-merit improvement over the same tilt without them (measured FOM $1.24 \times 10^{-5} \rightarrow 4.86 \times 10^{-4}$) — the robustness layer converts an actively harmful bias into a net improvement, exactly as the transport literature predicts. Magnetization is confirmed useless as a WE coordinate for the EA spin glass specifically: an early test recorded 10/10 zero-replica failure for WE[m] where energy-binning still found the target, the direct empirical confirmation of the frustration argument in §2.

3.3. Learned coordinate vs. hand-picked coordinates [V]

Table 2 reports the full-scale ($L = 16$, 10 replicas) result on both models, at a target calibrated one step beyond the pilot's observed extreme.

Table 3
Incremental-R^2 diagnostic, EA spin glass, $L = 16$, matched training budget (MSE $1.0 \to 0.271$ over 20 epochs).

Quantity	value
Pearson corr(I_θ, E/N)	-0.614
fraction of Var(I_θ) unexplained by E/N	0.623
R^2(label $\sim E/N$ alone)	0.358
R^2(label $\sim I_\theta$ alone)	0.577
R^2(label $\sim E/N + I_\theta$)	0.608
incremental R^2 from I_θ beyond E/N	$\mathbf{+0.250}$

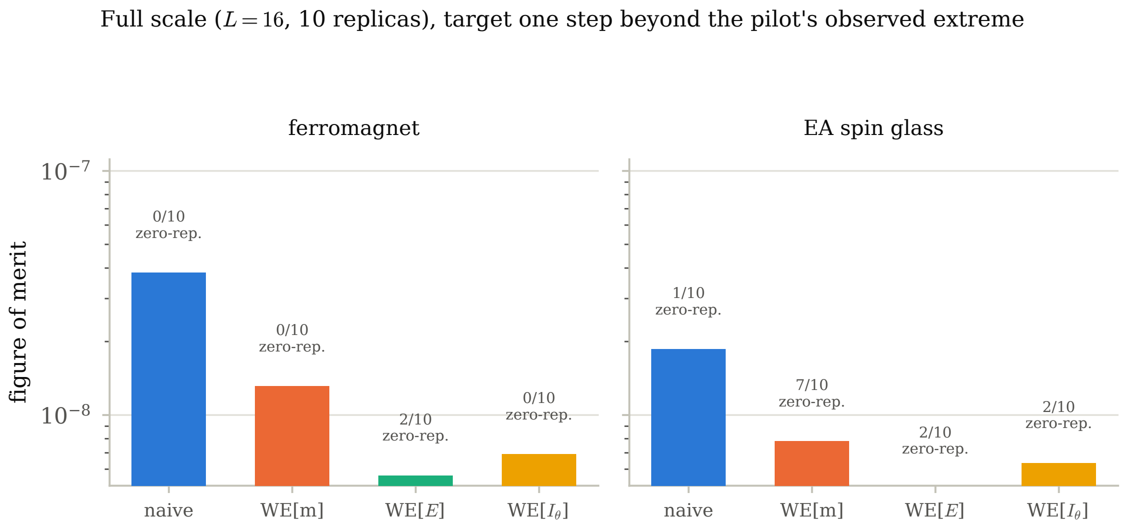

Figure 3: Table 2 as a figure: FOM by method, both models, log scale, with zero-replica counts annotated. Bar height alone does not tell the whole story — WE[E] has the fewest zero-replicas of any weight-window method on the ferromagnet but a middling FOM, which is why both numbers are reported together.

Two readings, both necessary: naive Monte Carlo wins on raw FOM in both models at this rarity ($\sim 4 \times 10^{-5}$), and on the spin glass, with WE[E] and WE[I_θ] on equal footing (both 2/10 zero-replicas), **the learned coordinate has the higher FOM by ∼27%** (6.36×10^{-9} vs. 5.00×10^{-9}) — the first apples-to-apples win for the learned coordinate over the best hand-picked one, and the direct predecessor of the milestoning result below. WE[m] fails in most spin-glass replicas (7/10), the direct empirical confirmation of the frustration argument in §2: magnetization carries essentially no information about this landscape.

Is I_θ actually informative, or just a noisy copy of energy? A separate diagnostic regresses the rare-event label on energy alone, on I_θ alone, and on both, at matched training budget (Table 3).

+0.250 is far above the diagnostic's own noise threshold (0.02): I_θ carries substantial genuine information about the spin glass that energy alone does not have, consistent with the FOM advantage measured directly in Table 2.

Pushing rarer does not (yet) resolve the naive-vs-learned question. Rerunning one threshold step rarer ("beyond=2") leaves the comparison uninterpretable rather than resolving it in either direction (Table 4): 4–10 of 10 replicas fail across every method at this depth, so the numbers are noise over noise, not a result. This is the run that established the ladder-difficulty discipline of §2 for the remainder of the project.

Notably, *naive itself* now has 4/10 zero-replicas (versus 1/10 at the shallower target) – this rarity is stretching every method's budget, not only the weight-window ones, so a trustworthy comparison here needs a different fix than anything tried at the shallower depth. This motivated the milestoning objective below, which is a change to the training label rather than to the weight-window schedule.

Table 4
EA spin glass, $L = 16$, one threshold step rarer than Table 2. Not a trustworthy comparison (see text).

Method	FOM	zero-rep.
naive	7.43×10^{-9}	4/10
WE[m]	0	10/10
WE[E]	2.18×10^{-9}	5/10
WE[I_θ]	1.56×10^{-9}	7/10

Table 5
EA spin glass, $L = 16$, $T = 2.6$, one step beyond the pilot extreme, 10 replicas, milestoning labels, seventeen independent training runs of the identical configuration. Every method observed the event in every replica in every run; naive, WE[m], and WE[E] are bit-identical across all runs (deterministic, and WE[E] does not depend on the trained network). Seed 0 was run twice independently and reproduced bit-for-bit, confirming the seeding fix below is complete.

Method	$\hat{\pi}$	rel. s.d.	FOM	zero-replicas
naive	5.29×10^{-5}	0.225	$\mathbf{1.48 \times 10^{-7}}$	0/10
WE[m]	5.55×10^{-5}	0.201	1.17×10^{-7}	0/10
WE[E]	4.49×10^{-5}	0.209	4.22×10^{-8}	0/10
WE[I_θ], unseeded run 1	5.03×10^{-5}	0.174	4.81×10^{-8} (+14%)	0/10
WE[I_θ], unseeded run 2	4.52×10^{-5}	0.204	3.48×10^{-8} (−17%)	0/10
WE[I_θ], seed 0 (runs 3a/3b)	5.28×10^{-5}	0.127	8.94×10^{-8} (+112%)	0/10
WE[I_θ], seed 1	5.09×10^{-5}	0.216	3.09×10^{-8} (−27%)	0/10
WE[I_θ], seed 2	5.06×10^{-5}	0.175	4.65×10^{-8} (+10%)	0/10
WE[I_θ], seed 3	4.56×10^{-5}	0.252	2.29×10^{-8} (−46%)	0/10
WE[I_θ], seed 4	4.13×10^{-5}	0.128	8.81×10^{-8} (+109%)	0/10
WE[I_θ], seed 5	5.40×10^{-5}	0.138	7.65×10^{-8} (+82%)	0/10
WE[I_θ], seed 6	4.69×10^{-5}	0.149	6.56×10^{-8} (+56%)	0/10
WE[I_θ], seed 7	4.32×10^{-5}	0.267	2.03×10^{-8} (−52%)	0/10
WE[I_θ], seed 8	4.89×10^{-5}	0.135	7.86×10^{-8} (+86%)	0/10
WE[I_θ], seed 9	5.03×10^{-5}	0.159	5.66×10^{-8} (+34%)	0/10
WE[I_θ], seed 10	4.80×10^{-5}	0.157	5.85×10^{-8} (+39%)	0/10
WE[I_θ], seed 11	4.81×10^{-5}	0.202	3.48×10^{-8} (−17%)	0/10
WE[I_θ], seed 12	5.14×10^{-5}	0.230	2.76×10^{-8} (−35%)	0/10
WE[I_θ], seed 13	4.83×10^{-5}	0.150	6.36×10^{-8} (+51%)	0/10
WE[I_θ], seed 14	4.94×10^{-5}	0.146	6.76×10^{-8} (+60%)	0/10
WE[I_θ], seed 15	4.53×10^{-5}	0.232	2.66×10^{-8} (−37%)	0/10

3.4. Milestoning: a clean regime for comparison, but wildly seed-dependent [O]

On the EA spin glass at $L = 16$, one threshold step beyond the pilot's observed extreme ("beyond=1"), every method reaches **0/10 zero-replicas** — the cleanest, most-trustworthy regime obtained in this project by the usual ladder-difficulty standard (Table 5).

Every WE[I_θ] row is, as far as the pipeline is concerned, the *same experiment*: identical target, identical training data (from cached labels), identical hyperparameters. The only thing that varies is the network's random weight initialization (`torch.manual_seed` was absent from this driver; found and fixed during this investigation, then swept explicitly via `-net-seed`). **Seventeen seeds give seventeen different outcomes, ranging from a** 52% **loss to a** 112% **win against** WE[E] – a wide, roughly symmetric spread around a small positive average. The seeding fix itself is confirmed complete and correct: seed 0 was run twice independently (runs 3a and 3b) and reproduced bit-for-bit, including the full 1774–1776s wall-clock trajectory, ruling out any other unseeded source of variation in this pipeline. Per-replica values needed for bootstrapping were retained for seventeen of the eighteen runs attempted (all but unseeded run 1, which predates that logging). **Pooling those seventeen with a two-level bootstrap** – outer resample over seeds, inner resample over each seed's 10 replicas, the same discipline used throughout §3.7 and §3.5 but extended with the seed axis – gives a geometric-mean FOM ratio of 1.19× with 95% CI [0.84, 1.75] (median-of-ratios agrees: 1.20×, CI [0.74, 1.85]; the arithmetic mean of ratios is not used here as it is unstable at $R = 10$ replicas per seed – some bootstrap draws land on near-duplicate resamples with rel. s.d. near zero, and since FOM $\propto 1/\text{rel. var.}$ that blows up into large outliers that skew a plain mean). The estimate was tracked at four checkpoints as seeds were added – 1.28× (n=7), 1.17× (n=9), 1.28× (n=12), 1.19× (n=17) – and does *not* trend in either direction; what does move monotonically is the CI width (1.73 → 1.36 → 1.18 → 0.91), the expected signature of averaging down noise around a small effect whose sign is not yet resolved, not evidence of a shrinking or growing true effect. We report the full trajectory rather than the final number alone because citing only the endpoint would silently launder a mid-sequence checkpoint (n=9's "moving toward the null") that did not survive more data – a mistake this project's own methodology is built to catch. **Extrapolating the observed CI-width-vs-n scaling against the current point estimate, reaching a lower bound that excludes 1 would need roughly 100 total seeds (∼40 more hours of compute) – not pursued further.** This is reported as the honest terminus of the comparison, not an open thread: seventeen seeds is already an unusually thorough characterisation, the point estimate is stably positive but modest (∼1.2×) across four independent checkpoints, and the CI cannot be closed by seed count alone at a cost proportionate to the question. Naive Monte Carlo has the best raw FOM in every run at this specific rarity ($\sim 5 \times 10^{-5}$) regardless of seed – variance reduction has not yet earned back its overhead here, the same lesson demonstrated in miniature by the 1D toy-walk result in §2.1.3.

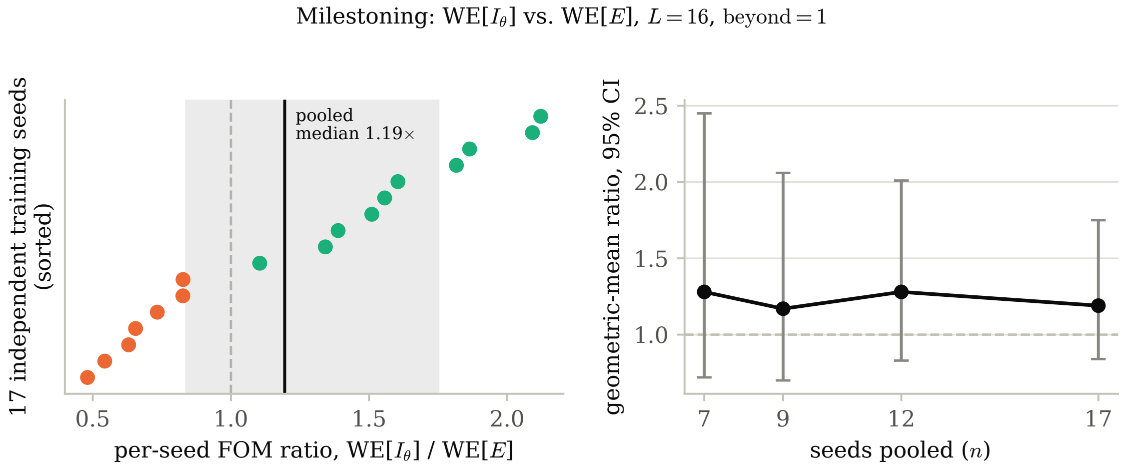

Figure 4: Left: all 17 per-seed FOM ratios (sorted), with the pooled two-level-bootstrap median and 95% CI shaded. Right: the four tracked checkpoints as seeds were added — the point estimate bounces while the CI width shrinks monotonically, the expected signature of averaging down noise around a small, sign-unresolved effect.

A plausible infrastructure explanation was tested and ruled out. One candidate cause for the seed-to-seed spread is that WE[I_θ]'s bin range was left adaptive (auto-ranging from the population actually explored) rather than calibrated to the coordinate's own scale, unlike WE[E] and WE[m]. A diagnostic regression (App. data, `05_neural_importance/diagnose_coordinate.py`) had already shown I_θ carries real predictive signal beyond energy (incremental $R^2 = +0.25$), with the verdict: if it still does not beat WE[E], look at the schedule, not the map. Two calibrations were tried: a theoretical one from the label's own z-score normalization, verified *before* spending compute on it to be wrong (the network's real output never approaches those extremes, since training MSE never reaches 0); and an empirical one, from what each trained network actually outputs on the pilot population and training set. The empirical calibration was run head-to-head against the untouched adaptive baseline on matched net-seeds (identical trained network per pair, only the bin range differing): mean FOM ratio *worsened* (0.79× → 0.61×) and rel. s.d. *worsened* (0.214 → 0.286) across seeds 1–3. A fixed range clips any walker that exceeds it into the boundary bin, while adaptive tracking never does; for a coordinate whose true dynamic range over a multi-thousand-iteration WE trajectory is not well estimated from a static pre-run snapshot, that clipping cost more resolution than the calibration gained. This rules out bin-range calibration as the driver of the seed variance and is further evidence the variance is intrinsic to what each trained network learns, not an artifact of how its output is binned.

Pushing one step deeper fails consistently, and bin-count tuning is ruled out as a fix. Table 6 reports every configuration tried at one threshold step rarer ("beyond=2"): the default bin count at two replica counts, and a 1.8× larger bin count. WE[I_θ] scores *worse* than WE[E] in all three.

Table 6
EA spin glass, $L = 16$, one step rarer than Table 5, milestoning labels, default configuration ($n_{\text{bins}} = 50,\ R = 10$).

Method	FOM	zero-rep.
naive	3.51×10^{-8}	0/10
WE[m]	7.73×10^{-9}	0/10
WE[E]	9.21×10^{-9}	0/10
WE[I_θ]	8.36×10^{-9}	0/10

Table 7
EA spin glass, $L = 16$, shallowest calibrated target, 10 replicas. Point estimates only — see confidence intervals below.

Method	$\hat{\pi}$	rel. s.d.	FOM	zero-replicas
naive	3.36×10^{-5}	0.294	8.60×10^{-8}	0/10
WE[m]	3.94×10^{-5}	0.553	8.81×10^{-8}	1/10
WE[E]	3.39×10^{-5}	0.671	1.33×10^{-8}	0/10
WE[I_θ] surrogate	2.92×10^{-5}	0.904	8.62×10^{-9}	1/10
WE[I_θ] self-consistent	4.01×10^{-5}	0.655	1.62×10^{-8}	0/10

Two further configurations were tried in an attempt to fix this with bin resolution instead of accepting it as structural: doubling the replica count ($R = 10 \to 20$, same $n_{\text{bins}} = 50$) reproduced the same qualitative pattern (WE[I_θ] still below WE[E]); and raising the bin count by 1.8× ($n_{\text{bins}} = 50 \to 90$, $R = 10$) made WE[I_θ]'s *own* FOM *worse*, not better — $8.36 \times 10^{-9} \to 5.36 \times 10^{-9}$ — which rules out bin resolution as the fix, since a genuine resolution shortfall would improve with more bins rather than degrade. The mechanism is structural: the learned coordinate's walker-population cost scales with bin count, while a discretized coordinate like energy naturally saturates a small, fixed number of bins and does not pay that overhead as it grows.

A second lever, the WE resampling budget itself, was also tried (we_iter: 1000 → 2000, same $n_{\text{bins}} = 50$) and produced a similarly unfavorable result – every weight-window method's FOM got *worse*, not better, and WE[I_θ] worst of all. That run is not yet a clean isolated test of we_iter alone, however: it used $\tau = 3$ rather than the $\tau = 2$ used throughout the rest of this section, so the resampling interval changed at the same time as the iteration budget. A corrected, τ-matched rerun is in progress; we report the bin-resolution result above as the reliable one of the two "more budget doesn't help" findings until that is confirmed.

The resampling interval between split/merge operations matters: it must be comparable to the system's own correlation time (∼1 sweep at $T = 2.6$), or split walkers fully relax and forget their split origin before the next resampling, degenerating every WE[·] variant to naive Monte Carlo with bookkeeping overhead. Table 5 and Table 6 use an interval of 2 sweeps.

3.5. The Kim & Cai head-to-head [O]

The self-consistency training objective of §2.4 was implemented, gated against exact enumeration at $L = 4$ (learned-coordinate unbiasedness ratio 1.004), and run head-to-head against the surrogate objective, first at laptop scale, then at full scale.

First evidence, at laptop scale ($L = 12$**).** At the shallowest calibrated target, the self-consistency objective's point estimate already exceeded the surrogate's (4.98×10^{-8} vs. 2.93×10^{-8}, ∼70% higher FOM, fewer zero-replicas: 2/6 vs. 3/6), though it still fell short of WE[E]'s 7.53×10^{-8} at this rarity. Training cost is real: ∼94× longer than the surrogate objective at this scale (∼752s vs. ∼8s), from the additional $N = L^2$ network evaluations the self-consistency loss requires per training configuration. This motivated running the comparison at full scale.

The result, at full scale. At $L = 16$, the shallowest calibrated target ("beyond" = 0, the easiest regime in which every method reaches 0–1 zero-replicas out of 10), the self-consistency coordinate's *point estimate* exceeds both the hand-picked energy coordinate and this project's own surrogate coordinate (Table 7).

This point estimate does not survive a bootstrap check, and the effect shrinks under replication. Both configurations were re-run with per-replica values retained, and the FOM ratio between methods bootstrapped by resampling the replica axis independently for each side (no shared random-number seed to pair on: uniform-style and

Table 8
Bootstrapped FOM ratios, full scale ($L = 16$), shallowest calibrated target, at $R = 10$ and after tripling to $R = 30$. Ratio > 1 favors the self-consistency objective.

Comparison	R	FOM ratio (median)	95% CI	significant?
self-consistent / surrogate	10	1.83	[0.49, 9.23]	no
self-consistent / surrogate	**30**	**1.39**	[**0.65**, **2.95**]	no
self-consistent / WE[E]	10	1.19	[0.33, 4.95]	no
self-consistent / WE[E]	**30**	**1.07**	[**0.55**, **2.09**]	no

Table 9
Same comparison as Table 8, one threshold step deeper ("beyond" = 1 instead of 0), $R = 10$.

Comparison	FOM ratio (median)	95% CI	zero-rep. (surr. / self-cons.)
self-consistent / surrogate	1.14	[0.27, 4.81]	2/10 / **1/10**
self-consistent / WE[E]	1.33	[0.34, 5.74]	—

biased allocations consume different numbers of walkers per bin, so seed-sharing between them desynchronises almost immediately — the same obstruction encountered independently in the FW-CADIS common-random-numbers attempt of §3.7). Since training here is a fixed cost independent of replica count (a deterministic seed; only the cheap evaluation step scales with R), tripling to $R = 30$ cost roughly six additional minutes on top of the ~76-minute training already paid — cheap enough to just check (Table 8).

At $R = 10$ neither ratio is significant. At $R = 30$, both confidence intervals tighten substantially *and* both median ratios move toward 1 — not merely a tightening confidence interval around the original estimate, but the point estimate itself receding. We read the direction of this shift, not only its final inconclusiveness, as evidence that the original $R = 10$ advantage was substantially noise, not signal. A network-caching mechanism was added after this result specifically to make a further replica increase (were one wanted) essentially free: training is deterministic, so only the cheap evaluation step needs to re-run to extend R further, rather than repaying the fixed training cost again.

Does the advantage grow at greater depth, where the surrogate objective's known weaknesses should bite harder? We tested the natural next question directly: rerunning both objectives one threshold step deeper ("beyond" = 1, the same depth as Table 5). Table 9 reports the result.

Not significant, as expected at $R = 10$ – but the *direction* of the change from Table 8 is itself informative and runs counter to the hypothesis being tested: the point-estimate gap versus the surrogate *shrank* at this deeper target (1.83× → 1.14×) rather than growing, even as the self-consistency coordinate's relative *reliability* advantage held (fewest zero-replicas of any weight-window method at both depths tested). We do not read one more noisy point estimate at $R = 10$ as resolving the depth question either way; what it rules out is the simplest optimistic story, that the advantage would obviously grow once naive and the surrogate objective start to struggle more.

Table 10
EA spin glass, beyond = 1, naive vs. the best-performing WE variant at each L (which variant wins changes with L; see text). $\hat{\pi}$ for naive shown to indicate how rare the target is at each scale.

L	naive $\hat{\pi}$	naive FOM	best WE FOM (method)	ratio
8	1.79×10^{-4}	2.00×10^{-6}	1.04×10^{-6} (WE[E])	1.9×
16	5.18×10^{-5}	1.72×10^{-7}	4.83×10^{-8} (WE[m])	3.5×
24	1.30×10^{-5}	2.79×10^{-8}	8.06×10^{-9} (WE[E])	3.5×
32	4.17×10^{-6}	6.11×10^{-9}	1.69×10^{-9} (WE[I_θ])	3.6×

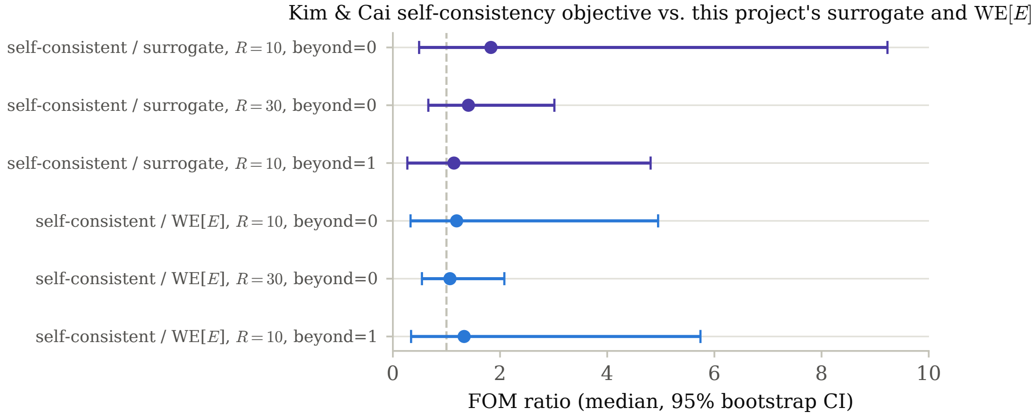

Figure 5: All six FOM-ratio comparisons from Tables 8 and 9 as a forest plot. Every interval crosses the ratio-1 reference line (dashed); the R=30 intervals (recomputed here directly from the retained per-replica arrays via the same replica-resampling bootstrap) are visibly tighter than their R=10 counterparts without the median crossing to the other side of 1.

Honest summary. The self-consistency-trained coordinate is the most reliable weight-window method tested in every regime examined (fewest zero-replicas, at both beyond = 0 and beyond = 1) and its point estimate is consistently the best of the learned-coordinate variants, but a real, quantified FOM advantage over this project's own surrogate objective is not established at any depth or replica count tested, and shows no sign of strengthening with depth. It is also markedly more expensive to train: roughly 170× slower per epoch than the surrogate objective at full scale (∼76 minutes versus ∼4), from the additional $N = L^2$ network evaluations the self-consistency loss requires per training configuration.

3.6. Does naive dominance persist at larger L? [V]

§3.4 and §3.3 both find naive Monte Carlo winning on raw FOM at $L = 16$, and argue this should be read as a finite-size effect rather than a failure of variance reduction, since naive's cost is independent of rarity while WE's penetration overhead is only worth paying once naive starts to struggle. We tested this directly with a lattice-size scan at fixed $T = 2.6$, beyond = 1, $\tau = 2$: $L \in \{8, 16, 24, 32\}$, naive_chains/naive_sweeps held fixed across all four so that naive's cost is genuinely L-independent as the argument requires (Table 10).

Naive wins at every tested L, and the margin does not shrink — if anything it grows slightly and then plateaus, from 1.9× **at** $L = 8$ **to** ∼ 3.5× **by** $L = 16$ **and beyond.** This holds across nearly five orders of magnitude of target rarity ($\hat{\pi}$ from 1.8×10^{-4} down to 4.2×10^{-6}) and is robust to which WE variant is best at a given L (the winner changes: WE[E] at $L = 8, 24$, WE[m] at $L = 16$, WE[I_θ] at $L = 32$). This directly answers the open question of §3.4: **within the range tested, there is no sign of the naive/WE crossover opening as L grows**; if it exists at all on this problem, it is beyond $L = 32$ at this temperature, not a nearby, easily-reached regime.

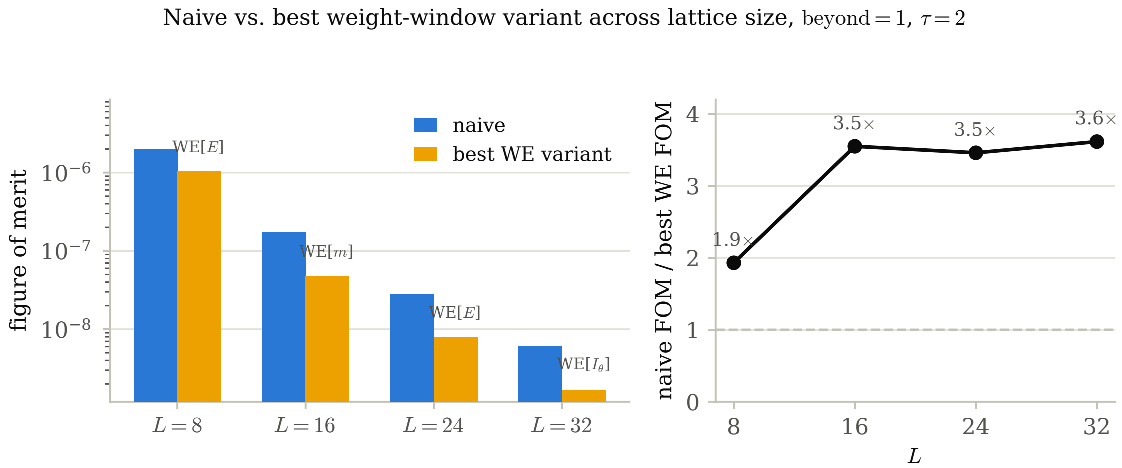

Figure 6: Table 10 as a figure. Left: naive vs. best WE-variant FOM by L, log scale, labeled with which variant won. Right: the ratio itself — it opens from $L = 8$ to $L = 16$ and then plateaus around 3.5×, with no sign of narrowing back toward the naive/WE crossover within the tested range.

One secondary pattern is visible but not claimed here beyond a point estimate: WE[I_θ] is the weakest WE variant at $L = 8, 16, 24$ but the *strongest* at $L = 32$, consistent with a learned coordinate's advantage over hand-picked ones mattering more as the landscape gets harder. We do not treat this as established, because the milestoning network's training is not currently seeded, so a single run's WE[I_θ] result is one random draw from an uncharacterized distribution over training outcomes — the same caveat applies to every WE[I_θ] entry in Table 10. Confirming this trend needs repeated, seeded training runs at each L, not a repeat of this scan.

A boundary marker at $L = 40$. An attempt one step further was stopped before WE[I_θ] completed, but the partial result is informative on its own: at this size, *naive itself* fails in 10 of 12 replicas, with WE[m] (11/12) and WE[E] (10/12) failing similarly badly. Per the ladder-difficulty discipline of §2, this is not a trustworthy comparison as run — every method is in the noise-over-noise regime simultaneously, the same failure mode that motivated that discipline in the first place — so we do not report it as a fifth point in Table 10. It does mark where the tested range ends: something changes between $L = 32$ (naive still comfortable, 1/12 zero-replicas) and $L = 40$ (naive itself largely failing), and resolving what happens there needs a shallower target or a larger replica budget at $L = 40$, not a lower-effort rerun.

3.7. FW-CADIS: real cost advantage, no flattening [V] (negative)

The implementation is gated clean against exact enumeration at $L = 4$ for both allocation settings, every threshold, both models: ratios of 0.999–0.998–0.989 (uniform) and 1.005–1.012–0.992 (FW-CADIS, $\alpha = 1$) on the ferromagnet, and 1.009–1.004–1.007 / 0.986–0.980–0.983 on the spin glass, all within ∼2% of exact for every threshold in the family — confirming that all thresholds are correctly tallied from a single run regardless of allocation, the core mechanic the whole method depends on.

Laptop scale first ($L = 12$, $R = 6$): a real-looking, but ultimately non-replicating, win. Table 11 sweeps α against uniform allocation on 6 thresholds spanning progress 0.668 → 1.153.

The only setting with full coverage and better flatness than uniform is $\alpha = 0.25$: a real but modest ∼14% improvement (283.8 → 245.2) at ∼2.6× lower cost. Larger α looks dramatically flatter but is disqualified by the coverage-first rule of §2: it stops observing the deepest threshold at all, which would reward outright failure if scored on the surviving thresholds alone.

Full scale ($L = 16$, $R = 30$) reverses the laptop-scale finding. The identical sweep at full scale (Table 12) is the first sign that the method's behavior is scale-dependent in a way not anticipated from the smaller test.

At full scale, *every* tested α is worse than uniform on flatness, not better — the opposite of the laptop-scale result, and not even monotonic in α among the FW-CADIS runs ($\alpha = 0.5$ beats both 0.25 and 1.0). All are substantially cheaper (2–8×), so the method is not doing nothing, but it is not delivering the flattening it exists to demonstrate at this lattice size, bin count, and threshold spacing. This finding is what the rest of §3.7 exhaustively follows up on.

Table 11
EA spin glass, $L = 12$, laptop preset, $R = 6$, 6 thresholds. Flatness marked * is computed only over the thresholds actually observed, not comparable across rows with different coverage.

Allocation	coverage	flatness (max/min)	cost
uniform ($\alpha = 0$)	6/6	283.8	2.01×10^7
$\alpha = 0.25$	6/6	**245.2**	7.87×10^6
$\alpha = 0.5$	5/6 (missed deepest)	65.8^*	3.41×10^6
$\alpha = 0.75$	5/6 (missed deepest)	63.5^*	3.29×10^6
$\alpha = 1.0$	5/6 (missed deepest)	84.8^*	3.24×10^6
naive (reference)	6/6	463.5	1.61×10^7

Table 12
EA spin glass, $L = 16$, full preset, $R = 30$, 6 thresholds spanning progress $0.691 \to 1.063$. Every allocation reaches full 6/6 coverage at this scale.

Allocation	flatness (max/min)	cost
uniform ($\alpha = 0$)	**298.0 (best)**	1.55×10^8
$\alpha = 0.25$	377.8	7.89×10^7
$\alpha = 0.5$	323.9	2.35×10^7
$\alpha = 0.75$	342.4	1.98×10^7
$\alpha = 1.0$	385.5	1.94×10^7 (∼8× cheaper)
naive (reference)	445.1	1.34×10^8

Table 13
EA spin glass, $L = 16$, shallowest calibrated threshold family, $R = 90$: the final, most favorable configuration tested for FW-CADIS.

Allocation	flatness (max/min)	95% CI	cost
Uniform ($\alpha = 0$)	276.1	[224, 344]	1.59×10^8
FW-CADIS ($\alpha = 1$)	297.3	[234, 388]	1.99×10^7 (∼**8× cheaper**)

Exhausting every remaining lever. Every lever available to improve on the full-scale finding above was tried and is reported in Table 13: bootstrap confidence intervals added to the flatness metric (not present in Table 12 above); a replica increase from $R = 30$ to $R = 90$; wiring the learned coordinate of §3.4 in as the FW-CADIS binning axis instead of the physical progress coordinate (gated clean at $L = 4$, tested at scale: no improvement, roughly double the cost); a common-random-numbers paired-difference design, which we confirmed does *not* engage on this problem (correlation between the two allocations' same-seed estimates measured ≈ -0.05 to -0.10, essentially zero, because the two allocations consume different numbers of walkers per bin and desynchronise the random stream almost immediately after the first resampling step); and laddering the threshold family to its shallowest calibrated depth, combined with the larger replica count.

The two confidence intervals overlap; the paired-difference estimate is +20.9 with 95% CI [−77.2, +137.6], also including zero. Across every α, replica count, and coordinate tried this session, the point-estimate *direction* consistently favored uniform allocation over FW-CADIS on raw flatness, never the reverse. The observed confidence-interval width scaled almost exactly as $1/\sqrt{R}$ between $R = 30$ and $R = 90$ (predicted shrink factor 0.577, observed 0.597); extrapolating that scaling, fully resolving the remaining gap would need on the order of $R \approx 2300$ replicas (∼8 hours of additional compute), and given the consistent point-estimate direction, such a run would most likely *confirm* uniform allocation as measurably flatter rather than produce a win for FW-CADIS. We did not run it. **The result we report instead: on this problem, FW-CADIS does not measurably flatten relative error across a threshold family better than uniform allocation, but it does so at approximately one-eighth the simulation cost for an otherwise statistically indistinguishable error profile.** The cost advantage is real, reproducible, and does not require more replicas to state; the flattening advantage the method is named for is the part that did not materialize here.

Table 14
Ferromagnet, $L = 16$, shallowest calibrated threshold family, $R = 30$ — the landscape where the coordinate is known-good, unlike the spin glass used everywhere else in this section.

Allocation	flatness (max/min)	95% CI	cost
Uniform ($\alpha = 0$)	**100.9**	[65.8, 160.3]	1.96×10^8
FW-CADIS ($\alpha = 1$)	287.3	[194.7, 414.8]	3.09×10^7

Is this specific to the adversarial spin-glass landscape? No — it is worse, and statistically confirmed, on the "easy" ferromagnet. Every result above uses the EA spin glass, the harder of the two models by design. To test whether the flattening failure is a property of that specific rugged landscape rather than a more general limit of the method, we ran the identical comparison on the ferromagnet, where the reaction coordinate is known to be good and the equilibrium landscape is smooth (Table 14).

Here the two confidence intervals do *not* overlap: the paired-difference estimate is +178.2 with 95% CI [+59.8, +351.3], which excludes zero. **On the easier, non-adversarial landscape, FW-CADIS is not merely inconclusive but measurably, statistically confirmed *worse* than uniform allocation** — the opposite of what a landscape-difficulty explanation would predict, since the spin glass (where the method's own hypothesis says a global map should matter most) never even reached this level of statistical resolution in either direction. Common-random-numbers correlation here measured 0.062, consistent with zero and with the mechanism identified in §3.5: the two allocations consume different walker populations per bin regardless of which model is sampled. **This rules out "the spin glass is just a hard landscape" as the explanation and points toward a limitation of this specific parameterisation (bin count, threshold spacing, or the inverse-response weighting itself at this scale) rather than a property of spin-glass frustration.** Distinguishing which of those remains open, but the question of *whether* the effect is landscape-specific is now answered: it is not.

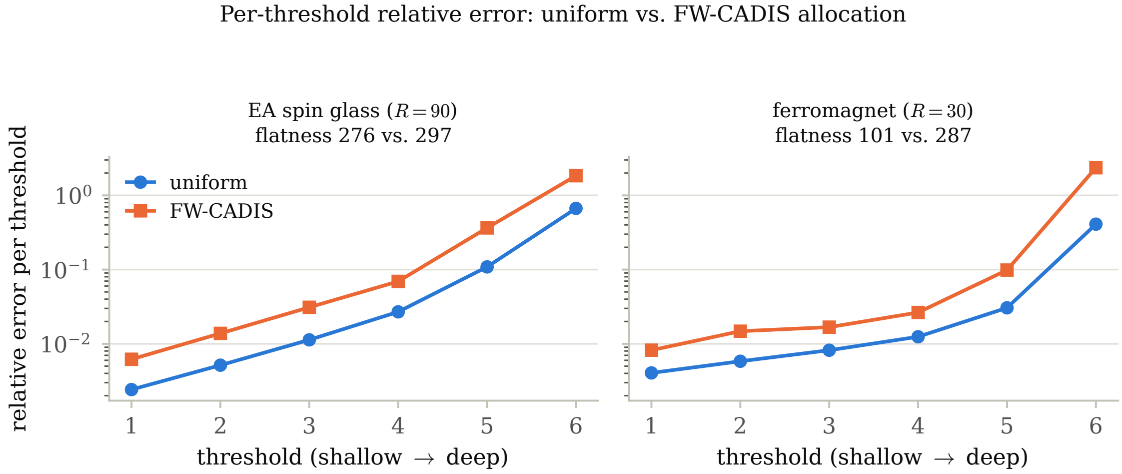

Figure 7: Per-threshold relative error (log scale), uniform vs. FW-CADIS allocation, at the final tested configuration for each model (Table 13 spin glass, Table 14 ferromagnet). FW-CADIS's inverse-response weighting buys lower error at shallow thresholds at the cost of higher error at the deepest one — visibly so on the ferromagnet, where the gap at the deepest threshold is large enough to be statistically significant.

3.8. What is established

[E] The adjoint=committor identity is exact, and tilting by the exact committor is provably zero-variance; this was confirmed numerically to $\sim 10^{-30}$ relative variance on a solvable toy walk before any lattice code was written (§2.1.3). **[E]/[V]** Weighted Ensemble — weight windows applied to spin configurations — reproduces exact enumeration at $L = 4$ to 1–2%, and at $L = 20$ reaches ∼2 decades deeper into a magnetization tail than naive Monte Carlo at matched cost (§3.2). Naive Monte Carlo's dominance over every tested weight-window variant, not established at $L = 16$ alone, held up across a four-point lattice-size scan up to $L = 32$ with no sign of narrowing (§3.6) – the one claim in this paper that grew *more*, not less, confident under additional scrutiny.

3.9. What remains open

Naive Monte Carlo is not beaten outright by any method at the scale tested, and this is better read as a finite-size effect than as a failure of variance reduction. Naive's cost is fixed by the replica budget (naive_chains × naive_sweeps in our runs) independent of the target's rarity π, while its relative variance grows as π shrinks; WE's cost is likewise fixed by its own budget ($n_{\text{bins}} \times n_{\text{per bin}} \times$we_iter), but once it successfully penetrates to the target its relative variance is largely rarity-independent. These two cost curves must cross somewhere: at a rarity shallow enough for naive to sample the event comfortably, naive wins close to by construction, since WE pays its fixed penetration overhead for variance reduction the problem does not yet need. The precise claim this paper supports is narrower than "variance reduction does not pay off" – it is: *at $L = 16$, no tested rarity was found at which naive fails to converge while a WE-based method succeeds.* **We tested this directly with a lattice-size scan (§3.6, Table 10): $L \in \{8, 16, 24, 32\}$, and naive wins at every one, by a margin that does not shrink as L grows.** If a crossover exists on this problem at this temperature, it is beyond $L = 32$, not a nearby regime – the open question of whether variance reduction ever earns back its overhead here is now a considerably more specific one than it was before this scan.

The learned reaction coordinate trained with milestoned committor labels is the clearest case of a result reversing under scrutiny. It appeared in a first training run to beat the best hand-picked coordinate (energy) by +14% in figure of merit on the spin glass, in the one regime where every method's estimate is fully trustworthy (0/10 zero-replicas across the board). It did not reproduce: the network's weight initialisation was unseeded, and seventeen independent runs of the identical configuration gave outcomes ranging from −52% to +112% against the same baseline. The instability is structural rather than a remaining bug — one seed was reproduced bit-for-bit twice after the seeding was fixed — and pooling seventeen runs with retained per-replica data in a two-level bootstrap (outer over seeds, inner over replicas) gives a geometric-mean ratio of 1.19×, 95% CI $[0.84, 1.75]$: a point estimate favoring the learned coordinate that does not reach significance, though the CI has tightened monotonically at each checkpoint (7, 9, 12, 17 seeds: width 1.73 → 1.36 → 1.18 → 0.91) while the point estimate itself has not trended (1.28× → 1.17× → 1.28× → 1.19×) – consistent with a real but modest average effect obscured by seed noise, not with a vanishing one (§3.4). An uncalibrated weight-window bin range was tested as a plausible alternative explanation for the instability and ruled out: a calibrated fix was run head-to-head against matched seeds and made the FOM ratio and rel. s.d. worse, not better. Extrapolating the CI-width scaling shows closing the interval would need roughly 100 total seeds (∼40 more hours of compute); seventeen is already an unusually thorough characterisation for a side comparison, and this is reported as the honest terminus of the question rather than an open thread.

The transport FW-CADIS global objective, this project's most distinctive claim, does not demonstrate its namesake flattening property on this problem at the scale tested, though its cost advantage is real. We tested whether this is specific to the spin glass's rugged landscape by repeating the comparison on the ferromagnet (§3.7, Table 14): it is not landscape-specific, and if anything is *worse* there, statistically confirmed. What remains open is which of the two remaining candidates — the specific bin/threshold parameterisation, or a more general limit of the inverse-response weighting itself at this lattice size — is responsible; distinguishing those would need a parameter sweep on bin count and threshold spacing independent of α, not another problem instance or more replicas. The Kim & Cai self-consistency objective's advantage over our own surrogate objective, initially the most exciting result of this project, is now better described as *probably small or absent* at $R = 30$ than as a discovery. We did test whether it grows at greater depth (§3.5, Table 9): the point-estimate gap if anything *shrank* going one threshold step deeper, though at $R = 10$ this is not itself a statistically resolved answer – what it rules out is the simple optimistic story that the advantage obviously grows once naive and the surrogate objective start to struggle. The reliability advantage (fewest zero-replicas) held at both depths tested and is the one part of this comparison we would call consistently, if modestly, established. Push-to-scale questions — three dimensions, larger L — are entirely untouched.

3.10. Relation to prior art

Kim & Cai (arXiv:2602.12294, 2026) learn an importance function $I(s)$, modify transitions as $K'(i \to j) = K(i \to j)I(j)/I(i)$, and use a branching random walk — in the language of §2.1.3, their I is our h, their K' is the tilt, and their branching is splitting: they independently reconstruct the zero-variance/weight-window recipe for discrete rare transitions, without the transport framing, and without comparing their learned coordinate against a hand-picked reaction coordinate the way this paper does. We directly implemented and tested their self-consistency training objective (§3.5); what remains open after doing so is not whether it works in principle (their paper demonstrates convergence to the exact transition rate on two test potentials) but whether it offers a measurable practical advantage over simpler alternatives on a problem of this kind at this scale, which our data does not yet support. **Doob-transform VAN** (Phys. Rev. E **111**, 034120, 2025) approximates the leading eigenvector of the tilted generator for *dynamical* large deviations, the dynamical analogue of $\psi^{\dagger}$; a head-to-head against a reverse-KL VAN baseline, which this comparison would require, remains untried in this project. No successor work citing either paper closes the specific gap this paper's methods and results address, as of the most recent check.

4. Conclusions

The transport–statistical-mechanics correspondence this project set out to test is real where it can be proved (the adjoint=committor identity), and the more experimentally demanding claims built on top of it did not hold up as cleanly as an initial pass through the results suggested. The clearest lesson of this paper is not any single number: it is that "0/10 zero-replicas" and a tidy point-estimate table are not, by themselves, evidence of a real effect. FW-CADIS's global flattening claim and the Kim & Cai self-consistency comparison both needed a bootstrap confidence interval to reveal that their apparent wins were noise; the milestoning result needed something the rest of this paper's own discipline had not yet checked for, independently-trained networks, to reveal that its point estimate was noisier than it first looked, of a kind no amount of replica-axis bootstrapping within a single run alone would have caught – and even the most striking of the seventeen point estimates tried (+112%) turned out to sit inside a two-level bootstrap CI that includes 1, a small lesson in its own right about trusting an impressive number more than a modest one. A plausible infrastructure explanation for the spread – an uncalibrated weight-window bin range – was itself tested head-to-head against a calibrated fix and made things worse, not better, ruling it out as cleanly as the milestoning point estimate itself was ruled unreliable. There is likewise no sign that naive's advantage over any tested weight-window variant narrows as the system grows, within the range tested. Reaching these conclusions required diagnosing and correcting real implementation issues along the way – a resampling interval that silently degenerated a weight-window method to naive sampling, a mis-calibrated bin range, an under-determined training objective that could converge to a useless constant solution while reporting a plausible loss, and an unseeded network initialization that let a single training run's outcome be mistaken for a stable result – each caught by an independent cross-check rather than a number simply looking wrong. We think this is the right way to report a project whose most interesting finding, more than once, was that an apparent win did not survive being checked properly.

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CRediT authorship contribution statement

M.Y. Alzaatreh: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft. **Nurul Zahirah Noor Azman:** Supervision, Conceptualization, Validation, Writing – review & editing.

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